Performance Analysis of OFDM under AWGN and Rayleigh Fading Channels Based on Software Defined Radio

Ivana Arum D[1], Rachmad Saptono[2], Muhammad Syirajjudin[3]
[1,2,3] Jaringan Telekomunikasi Digital, Politeknik Negeri Malang, Malang, Indonesia
Email: ivanaarumd@gmail.com, rachmad.saptono@polinema.ac.id, muh_syirajuddin@polinema.ac.id

***Abstract*— Orthogonal Frequency Division Multiplexing (OFDM) is a very efficient and popular modulation technique in modern wireless communication systems. OFDM offers advantages in terms of spectrum efficiency, resistance to multi-path interference, and is able to support high data rates. This OFDM system uses the concept of Software Defined Radio (SDR) because it is able to implement directly with the help of USRP B210 hardware. The OFDM system consists of a transmitter and receiver that are processed by digital signal processing to encode, decode, and modulate data. This study uses GNU Radio software as a transmitter and receiver model. The purpose of this study is to compare the performance of real-time data transmission through the AWGN (Additive White Gaussian Noise) and Rayleigh Fading channels. The results of the study show that the performance of the OFDM system built, namely the use of BPSK modulation on the AWGN channel has the best performance with the lowest BER results of 0.376 and SNR 14 dB.**

***Index Terms*— GNURadio, OFDM, SISO, USRP**

I. INTRODUCTION

The development of technology in the field of telecommunications has progressed very rapidly. Telecommunications tools consisting of devices and applications used continue to change following the development of telecommunications technology. The development of communication technology, the demand for data rate services that are getting bigger and must be fast through wireless communication is also increasing [1].

To achieve a higher data-rate, of course, requires a larger bandwidth in single carrier transmission due to the minimum bandwidth. When the signal bandwidth becomes larger than the coherent bandwidth in the wireless channel, the signal will be exposed to multi-path fading resulting in Inter-Symbol Interference (ISI) in the wireless channel. To overcome this problem, effective and efficient multicarrier transmission is needed, namely Orthogonal Frequency Devision Multiplexing (OFDM)[2]. OFDM is one of the transmission techniques with several subcarrier frequencies that are mutually orthogonal. Subcarriers will overlap each other without interference[3].

The advantage of OFDM is that it has a high capacity and is resistant to fading, but has the disadvantage of being vulnerable to Carrier Frequency Offset (CFO) caused by the Doppler effect[4]. In addition to cellular implementations such as WiMAX and LTE, and digital broadcasting standards such as DVB-T, OFDM is also used in various IEEE 802.11 standards, including various versions of wifi [5].

Communication systems using hardware require high time, effort, and development costs. SDR techniques provide a more efficient approach for researchers, reducing the time, effort, and cost required to implement wireless communication systems. The main goal of SDR is to transform radio hardware challenges into software-based solutions[6]. SDR techniques have hardware including USRP has powerful data processing capabilities to accurately process wireless data in real time and provides an efficient hardware platform for radio software research [7]. Several platform variations, such as GNU Radio, MATLAB, SIMULINK, and LABVIEW, are used in the communications field to work with and evaluate OFDM performance. [8].

GNU Radio is an open platform (free software for building SDRs, written in the Python programming language) and open Universal Software Radio Peripheral (USRP) hardware and software developed by Ettus Research, respectively. The platform is designed for run-time signal processing [9].

This research analyzes OFDM using Single Input Single Output (SISO) technique using AWGN and Rayleigh fading channel comparison.

II. THEORY

A. OFDM

Orthogonal Frequency Division Multiplexing (OFDM) is a transmission technique that uses several frequencies that are mutually perpendicular (orthogonal).

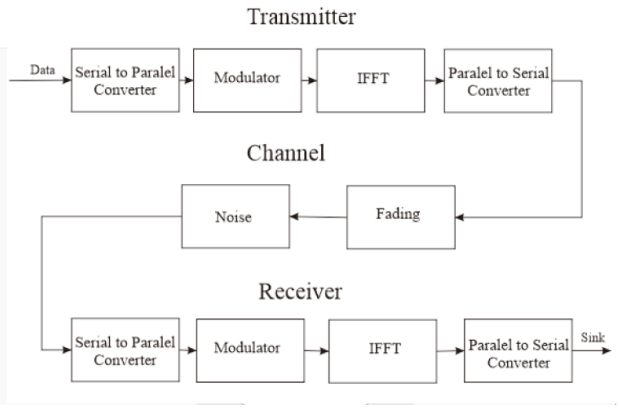


Figure 1. Block Diagram of OFDM System [10]

OFDM is applied to a signal that has been modulated, thus second-level modulation. The trick is to divide the data in parallel on a number of narrow-band subchannels,

then each data on the subchannel is modulated with a carrier subfrequency that is mutually orthogonal, then transmitted simultaneously. With this orthogonal nature, the sub-carriers can be overlapped without causing Inter Carrier Interference (ICI) effects. The following OFDM block diagram can be seen in Figure 1.

B. AWGN Channel

AWGN stands for Additive White Gaussian Noise. In communication, an AWGN channel is a channel where information is given interference in the form of linear addition of white noise with constant spectrum density and Gaussian distribution of noise samples.

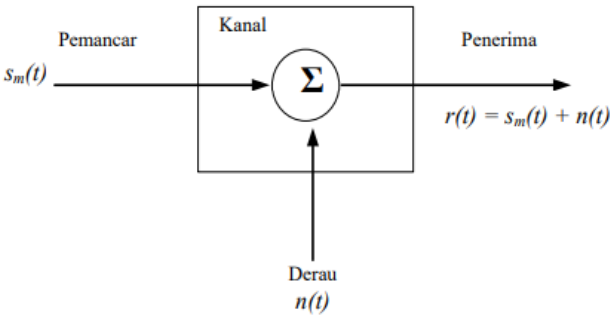


Figure 2. AWGN Channel Model [3]

Figure 2 is an AWGN channel model. If the transmitted signal Sm(t), the channel will be affected by noise n(t) so that the received signal r(t) becomes:

$$r(t) = S_m(t) + n(t) \quad (1)$$

C. Rayleigh Fading Channel

In mobile communication systems, signal propagation between the transmitter and receiver through different paths results in varying received signal strength. Due to the low antenna of the receiver and the presence of building structures surrounding the receiver, it causes rapid fluctuations in the summation of multipath signals. The fluctuations in signal amplitude that occur are random and cannot be determined in advance how large and when they occur. According to statistical distribution, this is known as the Rayleigh distribution. Rayleigh distribution is one of the distributions that can be modeled to represent fading. Rayleigh fading occurs when there is no dominant Line Of Sight (LOS) path between the sender and receiver[11].

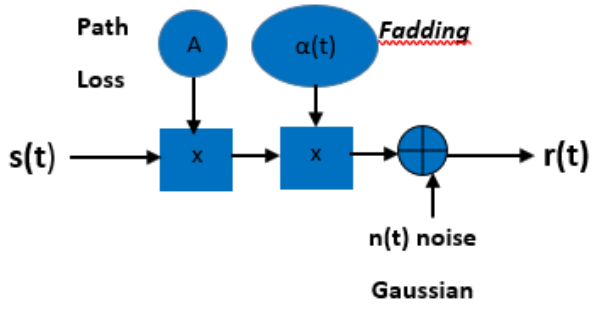


Figure 3. Rayleigh Channel Model

The channel response is affected by the random phase between the delay and the random Doppler effect and is affected by the random amplitude of the propagation path.

The channel response can be expressed by equation 2:

$$h(t,\tau) = \sum_{l=0}^{L} r^l(t)e - j(\omega t - \varphi D)\delta(t,\tau) \quad (2)$$

The received signal from the channel response can be written as equation 3:

$$y(t) = re^{j\varphi}.s(t) + n(t) \quad (3)$$

r is the Rayleigh distribution or the values of the impulse response with respect to time.

III. System Design

In the system design stage, an OFDM implementation is designed using GNU Radio software and USRP B210.

A. Transmitter

In the GNU Radio transmitter block in Figure 4, the 'Source File' reads the text file data that will be forwarded to the source address specified in the parameter file block. The 'Variables' block is used for initialization of parameters like sampling rate, FFT length, roll rate, etc. We can use variables in the initialization of other block parameters. 'Virtual sink' and 'Virtual source' are like connectors between blocks. For better representation, we divide the flowchart into sections using virtual Source and sink blocks. The 'Chunks to Symbols' block converts packed or unpacked data into a stream of float or complex symbols.

'Stream to tagged stream' block. increments the length tag with data depending on the packet length and locks the length tag in regular intervals. 'Stream CRC32' is used for error detection. This block adds a 32 bit CRC to all incoming packets. Depending on the header for material and header length, 'Packet header generator' generates the header. 'Tagged Stream Mux' takes an 'N' tagged stream as input and multiplexes it. The output stream has the same modified length tag with the addition of length tag 'N'. Here, the transmitter header bits are multiplexed with the payload bits using this block.

The 'OFDM carrier Allocator' assigns a unique carrier to each symbol depending on the polynomials used in objects like occupied carrier and pilot. The 'Fast Fourier transform' (FFT) block performs the reverse FFT. This is used as IFFT in OFDM transmitters. The OFDM symbol and guard interval are attached with the OFDM cyclic prefix. The 'USRP sink' block selects the SDR device to transmit data over the channel using the device address parameters. It is used to stream samples to USRP B210 which acts as a transmitter.

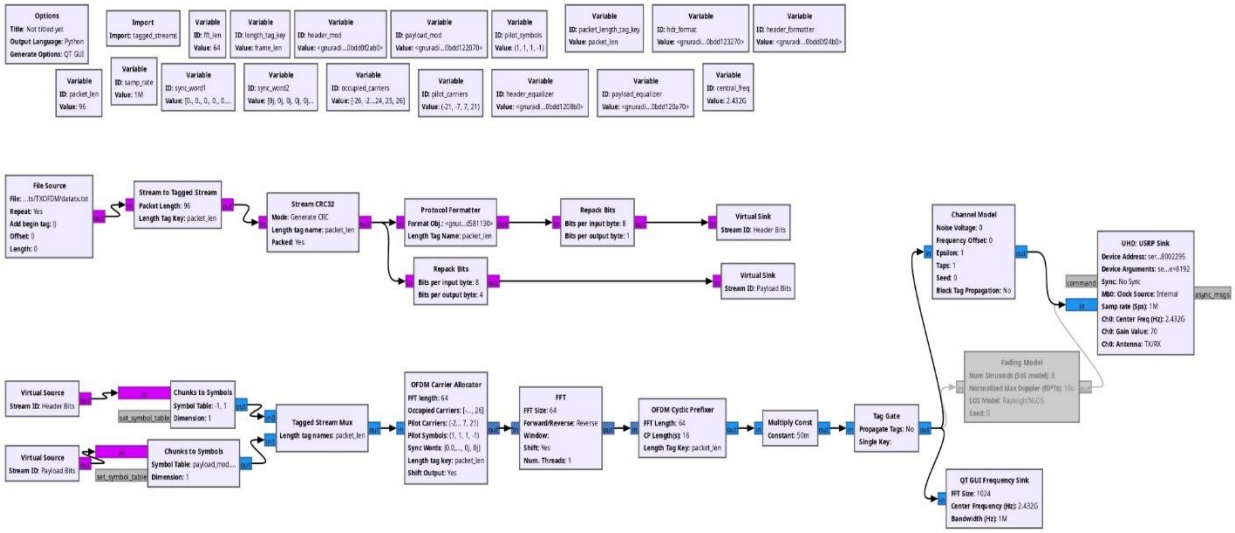


Figure 4. GNU Radio Transmitter Block

B. Receiver

Gambar 5 Figure 5 is the GNU Radio Receiver block. From the transmitter capture, the receiver process continues to receive the signal through the block, 'UHD Source USRP' receives the digital signal sent through the channel. MPSK SNR Estimator Probe to input the SNR process. Then from the UHD USRP source also proceeds to the 'Schmidt Cox Synchronization' Block used to eliminate symbol and frequency offsets. The 'Delay' block delays the OFDM symbols with a fixed number of samples. 'Header or Payload Demux' separates the header section as per the demodulated header and provides the data for demodulation.

The 'Packet Header parser' block is the reverse block of the 'Packet header generator'. Its header object must be the same as the header generator. The 'Repack Bits' block increases the overall number of bits according to the parameter length tag. At the receiver, the 'FFT' block is used as forward FFT as well as to synchronize the OFDM demodulator. 'OFDM channel estimation' gives the data symbols after removing the synchronization symbols. This is used to determine the offset frequency. 'OFDM frame equalizer' removes the line frequency offset present in the OFDM symbols. 'OFDM serializer' gives us complex symbols. The file results after undergoing the receiver process out a crc32 stream which is continued to the file sink. The BER block will compare the received file with the sent one.

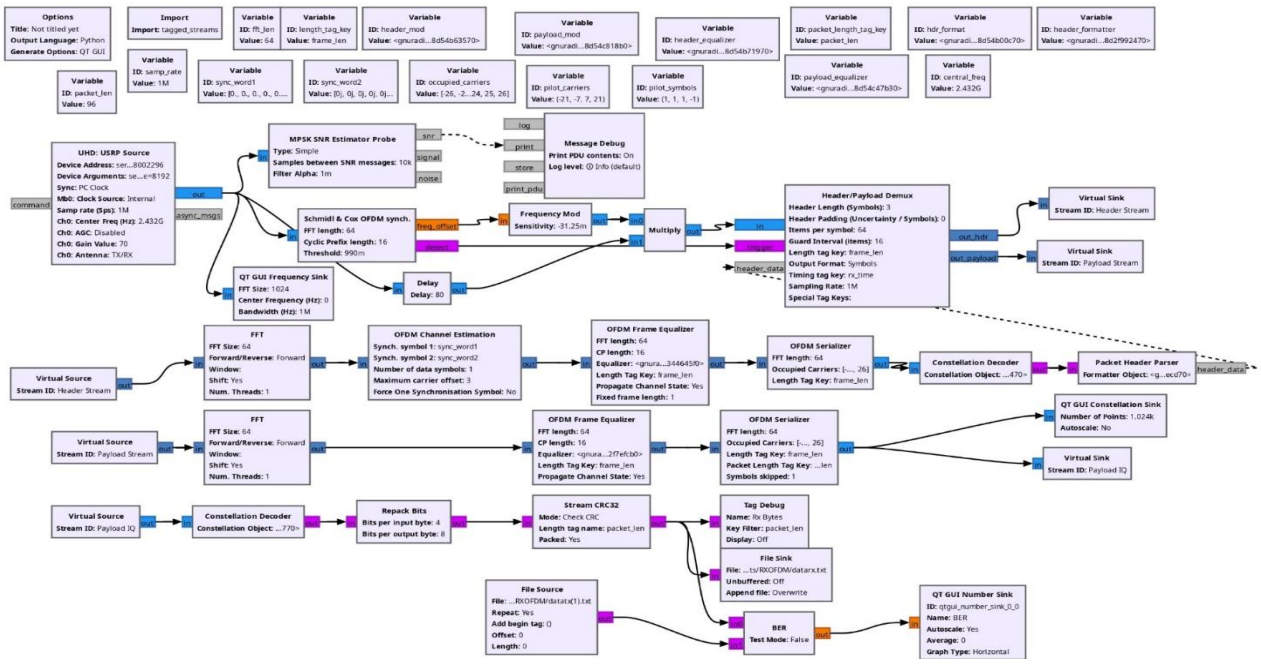


Figure 5. GNU Radio Receiver Block

IV. IMPLEMENTATION

This research uses GNU Radio software connected to the USRP tool. All parameter values will be set on the software, both transmitter and receiver.

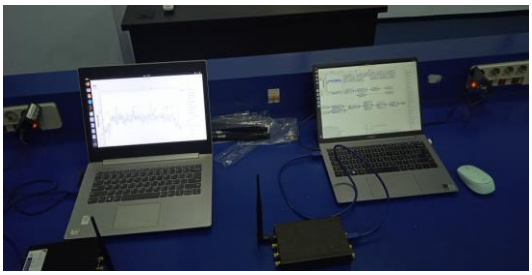


Figure 6. Connecting USRP to a Laptop with GNU Radio and USRP Drivers Installed

The OFDM implementation that was built has parameter values in Table 1.

TABLE I
PARAMETER VALUE

Parameter Sistem	Nilai
Center Frequency	2,432 GHz
FFT/IFFT *Size*	64
Cyclic Prefix Length	16
Modulation	BPSK, QPSK, 16-QAM
Tx Gain	70 dB
Rx Gain	70 dB

The following in Table 2 are the hardware and software specifications:

TABLE II
HARDWARE SOFTWARE SPECIFICATIONS

Transmitter	
Device	Advan
Processor	Intel(R) Core(TM) i5-1035g7 cpu 120 GHz 1,50 GHz
RAM	8 GB
OS	Ubuntu 24.04
Type GNU Radio	3.10.9.2
Receiver	
Device	Lenovo
Processor	AMD A9-9425 RADEON R5, 5 COMPUTE CORES 2C+3G 3.10 GHz
RAM	8 GB
OS	Ubuntu 24.04
Type GNU Radio	3.10.9.2

V. RESULTS AND ANALYSIS

This research uses GNU Radio software connected to the USRP tool. All parameter values will be set in the software, both transmitter and receiver. The OFDM spectrum is displayed on the transmitter side in Figure 7. The resulting signal forms a subcarrier by following the inputted FFT length. The signal has a relative gain with an amplitude of -40 dB. Figure 8 is the OFDM spectrum displayed on the receiver side with an SNR result of 11 dB. This signal shifts the relative gain position to an amplitude of -30 dB.

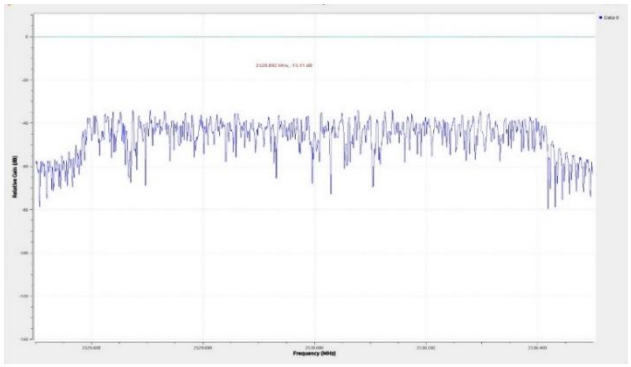


Figure 7. Transmitter Results

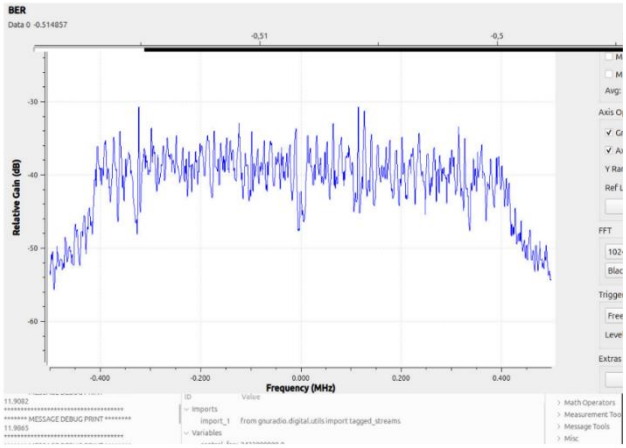


Figure 8. Receiver Results

Constellation is a graphical representation of modulation symbols in the complex plane. Each point in the constellation diagram represents one possible symbol of the modulation used. The types of modulation used in this study are BPSK, QPSK, and 16-QAM. Each modulation technique has a different constellation diagram. BPSK has 2 constellation points as in Figure 9, QPSK has 4 constellation points as in Figure 10, and 16-QAM has 16 constellation points as in Figure 11.

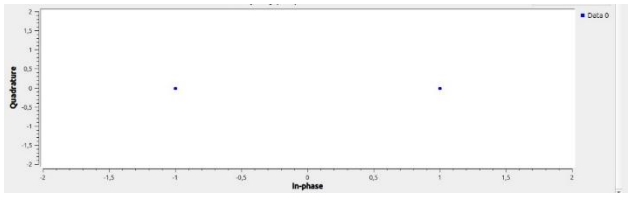


Figure 9. BPSK Constellation

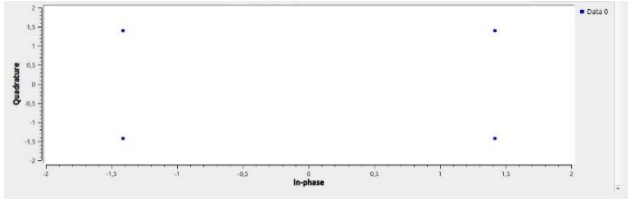


Figure 10. QPSK Constellation

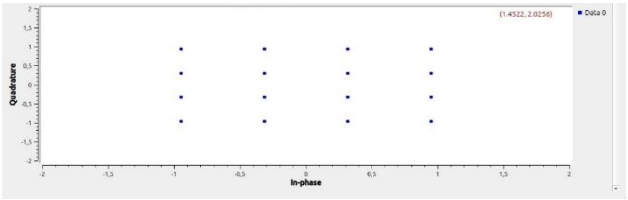


Figure 11. 16-QAM Constellation

The OFDM system built in this study uses different types of channels, namely AWGN and Rayleigh fading. To set the AWGN channel in GNU Radio is called "Channel Model" while setting the Rayleigh fading channel is called "fading model". During testing, both channels are used, one of which is if the channel model is activated then the fading model is deactivated and vice versa if the fading model is activated then the channel model is deactivated. In addition, the GNU Radio circuit also regulates the type of modulation technique by using the header payload modulation, namely BPSK, QPSK, and 16-QAM.

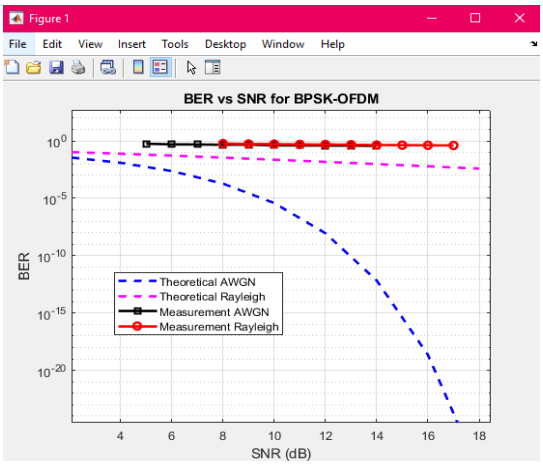


Figure 12. BER to SNR BPSK Graph

Figure 12 is a BER to SNR graph using BPSK modulation consisting of theoretical results and SDR measurement results. Based on AWGN theory, it has a BER level value ranging from 10-3 to 10-25 and an SNR level of 2 dB to 17 dB. In the SDR measurement results, the AWGN channel has a BER level ranging from 10-0,376 to 10-0,578 and has an SNR level value of 5 dB to 14 dB. Based on Rayleigh theory, it has a BER level value ranging from 10-2 to 10-3 and continues while the SNR level is from 2 dB to 18 dB and continues. In the SDR measurement results, the Rayleigh channel has a BER level ranging from 10-0,409 to 10-0,589 and has an SNR level value of 8 dB to 17 dB.

Figure 13 is a BER to SNR graph using QPSK modulation consisting of theoretical results and SDR measurement results. Based on the AWGN theory, it has a BER level value ranging from 10-3 to 10-25 and still continuing while the SNR level is from 2 dB to 18 dB and still continuing. In the SDR measurement results, the AWGN channel has a BER level ranging from 10-0,380 to 10-0,588 and has an SNR level value of 6 dB to 15 dB. Based on Rayleigh theory, it has a BER level value ranging from 10-2 to 10-3 and still continuing while the SNR level is from 2 dB to 18 dB and still continuing. In the SDR measurement results, the Rayleigh channel has a BER level ranging from 10-0,390 to 10-0,591 and has an SNR level value of 9 dB to 18 dB.

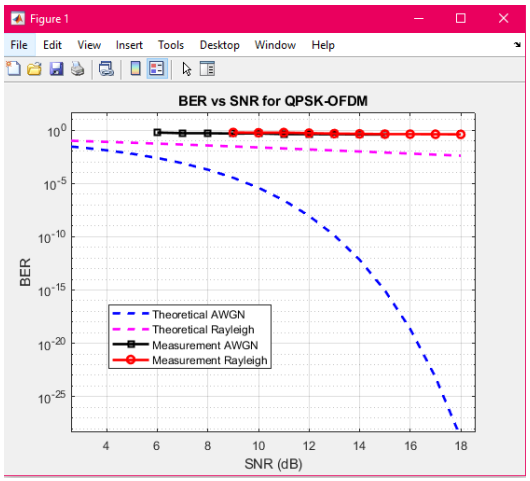


Figure 13. BER to SNR QPSK Graph

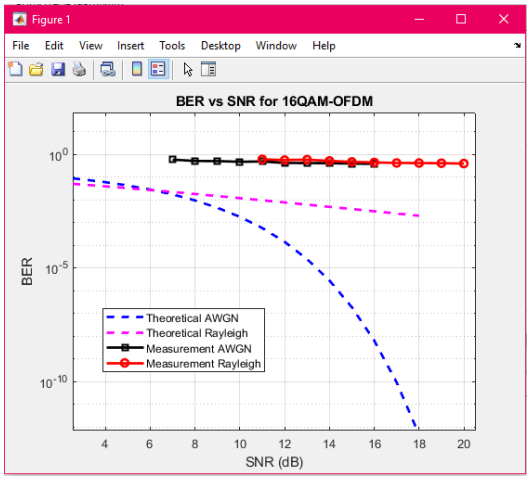


Figure 14. BER to SNR 16-QAM Graph

Figure 14 is a BER to SNR graph using QPSK modulation consisting of theoretical results and SDR measurement results. Based on AWGN theory, it has a BER level value ranging from 10-1 to 10-12 and still continuing and an SNR level of 2 dB to 17 dB and still continuing. In the SDR measurement results, the AWGN channel has a BER level ranging from 10-0,389 to 10-0,578 and has an SNR level value of 7 dB to 16 dB. Based on Rayleigh theory, it has a BER level value ranging from 10-2 to 10-3 and still continuing while the SNR level is from 2 dB to 18 dB and still continuing. In the SDR measurement results, the Rayleigh channel has a BER level ranging from 10-0,401 to 10-0,599 and has an SNR level value of 11 dB to 20 dB. The results of the AWGN theory have a BER level that varies from 10-3 to 10-25, which means that the BER value reduces the error rate faster. Meanwhile, when compared to the results of the measurement, BPSK modulation has a constant BER level ranging from 10-0,376 to 10-0,578 so that the measurement line is monotonically straight. Because the BER value that only changes slightly still has a large error

rate value. When viewed from the results of the AWGN SNR, theoretically it has a value of up to 17 dB while the measurement results have a value of up to 14 dB. The results of the Rayleigh fading theory have a more constant BER level but a relatively large BER level so that it takes longer to decrease than AWGN, namely from 10-1 to 10-3 and still continuing. This is in accordance with the Rayleigh theory which has more complex channel interference with multipath fading effects so that it is increasingly susceptible to interference which causes a relatively large error rate. Meanwhile, from the results of SDR measurements, the BER results are also relatively constant, ranging from 10-0,409 to 10-0,589 , which means that the value also has a high error rate. When viewed from the SNR Rayleigh results, theoretically it has an SNR ranging from 2 dB to 18 dB and above. While the measurement results have an SNR level ranging from 8 dB to 17 dB.

In the difference in theoretical results and SDR measurements, each modulation (BPSK, QPSK, and 16-QAM) both on the AWGN and Rayleigh fading channels. The BER results from the measurements are still relatively large, around 10-0,3 to 10-0,5 when compared to the theoretical results. This transmission is carried out continuously so that the data continues to run the sending process which causes the data sent to be increasingly different from the data received which causes a high BER value. This difference is also caused by direct implementation, it can be affected by other interference that increases bit errors. In SDR implementation, time, frequency, and phase synchronization can be more difficult to do than ideal conditions in theory. This imperfection in synchronization can lead to more errors which causes the BER value to increase.

VI. CONCLUSION

The conclusion of this study can be summarized below:

1. The OFDM SISO system built using SDR devices, namely GNU Radio and USRP B210, can function well in the transmission process between the two. The results of the measurement of relative gain or spectrum amplitude are -40 dB from the transmitter side to -30 dB from the receiver side.
2. In the AWGN channel, the BPSK modulation technique shows the best performance with a BER value of 0.376 with an SNR of 14 dB. While QPSK modulation has a higher BER of 0.380 with an SNR of 15 dB, and 16-QAM modulation has a higher BER than BPSK and QPSK, namely 0.389 with an SNR of 16 dB.
3. In the Rayleigh fading channel for BPSK, QPSK, and 16-QAM modulation, the BER value is higher than the AWGN channel. This shows that the effect of multipath fading interference can affect system performance.
4. The greater the SNR value, the smaller the BER value.
5. The further the distance between the transmitter and receiver, the lower the SNR value. This is related to the increasing BER value so that the received text file data has more lost and random words.

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